

Android - Java

Listview

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Outline

1 Basic Listview

2 Advanced Listview

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```
1 <ListView
2   android:id="@+id/nameListview"
3   android:layout_width="match_parent"
4   android:layout_height="wrap_content"
5   app:layout_constraintEnd_toEndOf="parent"
6   app:layout_constraintStart_toStartOf="parent"
7   app:layout_constraintTop_toTopOf="parent" />
```

Listing 1: Add listview component to activity.xml

Datasource and adapter

```
1 private String[] names = {"Nguyen Van A", "Tran Van B", "Nguyen Ho Van  
   D"};  
2  
3 ListView nameListview = (ListView) findViewById(R.id.nameListview);  
4 ArrayAdapter<String> arrayAdapter = new ArrayAdapter<String>(  
  (getApplicationContext(),  
5       android.R.layout.simple_list_item_1, names);  
6 nameListview.setAdapter(arrayAdapter);
```

Listing 2: Link data source to listview by using adapter

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Listview item

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <androidx.constraintlayout.widget.ConstraintLayout xmlns:app="http://
  schemas.android.com/apk/res-auto"
3   xmlns:tools="http://schemas.android.com/tools"
4   xmlns:android="http://schemas.android.com/apk/res/android"
5   android:layout_height="wrap_content"
6   android:layout_width="match_parent"
7   android:paddingTop="20dp"
8   android:paddingBottom="20dp">
9
10  <TextView
11    android:id="@+id/name"
12    android:layout_width="wrap_content"
13    android:layout_height="match_parent"
14    android:layout_marginStart="10dp"
15    android:text="TextView"
16    android:textAlignment="textStart"
17    app:layout_constraintBottom_toBottomOf="parent"
18    app:layout_constraintStart_toStartOf="parent"
19    app:layout_constraintTop_toTopOf="parent" />
20
21  <TextView
22    android:id="@+id/dob"
23    android:layout_width="wrap_content"
24    android:layout_height="match_parent"
25    android:layout_marginEnd="10dp"
26    android:text="TextView"
27    android:textAlignment="textEnd"
28    app:layout_constraintBottom_toBottomOf="parent"
29    app:layout_constraintEnd_toEndOf="parent"
30    app:layout_constraintTop_toTopOf="parent" />
31 </androidx.constraintlayout.widget.ConstraintLayout>
```

Listing 3: Prepare an item view

Datasource

```
1
2 class Student{
3     private String name;
4     private int dob;
5
6     public Student(String name, int dob){
7         this.name = name;
8         this.dob = dob;
9     }
10
11    public String getName() {
12        return name;
13    }
14
15    public void setName(String name) {
16        this.name = name;
17    }
18
19    public int getDoB() {
20        return dob;
21    }
22
23    public void setDoB(int dob) {
24        this.dob = dob;
25    }
26 }
```

Customized Adapter

```
1  class StudentListAdapter extends ArrayAdapter<Student>{
2      int resource;
3      private List<Student> students;
4      public StudentListAdapter(Context context, int resource, List<
5      Student> students) {
6          super(context, resource, students);
7          this.students = students;
8          this.resource = resource;
9      }
10     @NonNull
11     @Override
12     public View getView(int position, @Nullable View convertView,
13     @NonNull ViewGroup parent) {
14         View v = convertView;
15         if (v == null){
16             LayoutInflater vi;
17             vi = LayoutInflater.from(this.getContext());
18             v = vi.inflate(this.resource,null);
19         }
20         Student s = getItem(position);
21         if (s!=null){
22             TextView nameTextView = (TextView) v.findViewById(R.id.
23             name);
24             TextView dobTextView = (TextView) v.findViewById(R.id.dob)
25             ;
26             if(nameTextView!=null)
27                 nameTextView.setText(s.getName());
28             if(dobTextView!=null)
29                 dobTextView.setText(String.valueOf(s.getDoB()));
30         }
31         return v;
32     }
33 }
```

Link datasource to adapter

```
1 public class MainActivity extends AppCompatActivity {
2     @Override
3     protected void onCreate(Bundle savedInstanceState) {
4         super.onCreate(savedInstanceState);
5         setContentView(R.layout.activity_main);
6
7         List<Student> students = new ArrayList<Student>();
8         students.add(new Student("Nguyen Van A", 1999));
9         students.add(new Student("Nguyen Tan B", 2000));
10        students.add(new Student("Tran Xuan B", 2001));
11
12        ListView list = (ListView) findViewById(R.id.nameListview);
13        StudentListAdapter adapter = new StudentListAdapter(
14            getApplicationContext(), R.layout.list_item_student, students);
15        list.setAdapter(adapter);
16    }
```